Classical Field Theory — Exercise Sheet 8

Slices, boundary conditions and causal brackets

Hints and answers

For real Klein–Gordon variations with compact smooth Cauchy data, use $\omega^\mu(h_1, h_2) = h_1\partial^\mu h_2 - h_2\partial^\mu h_1$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. All fields below are smooth. The last problem explicitly restricts the field equation to the massless wave in $3+1$ dimensions.

Exercise 8.1 (An unbounded Cauchy graph). Let $\tau(x) = t_* + v \cdot x$ on $\mathbb{R}^d$, where $t_* \in \mathbb{R}$ and $v \in \mathbb{R}^d$, $|v| < 1$. Let $h_1, h_2 \in \mathcal{S}_m$ be the actual compact-Cauchy-data solutions of Lecture 8, $m \geq 0$.

(a) Prove that the graph is spacelike and meets each inextendible timelike curve exactly once. A future timelike curve can be parametrized by coordinate time with $|\dot{x}(t)| < 1$; justify that its time range is all of $\mathbb{R}$. Here an inextendible curve has no past or future endpoint in Minkowski spacetime.

(b) Although τ is unbounded, prove that the current is supported in a compact subset of the region between the graph and $t = 0$. Give an explicit spatial bound in terms of t_*, $|v|$, and a radius R containing both sets of initial data. Use signed time integration to prove $\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_0(h_1, h_2)$, including where the graph crosses zero.

(c) Choose nonzero real $f \in C_c^\infty(\mathbb{R}^d)$ and solutions with data $(f, 0)$ and $(0, f)$. Determine their pairing on this graph.

For comparison, the following general result is supplied: in full Minkowski spacetime, the same compact-data solutions have finite, equal symplectic integrals on every smooth spacelike Cauchy surface with future orientation. It uses spacelike compact support, not just current closure. See Khavkine, §3.3.3, Lemma 3.11, p. 41, arXiv:1402.1282. Your proof here should establish the affine case directly.

Hint. Along a timelike curve differentiate $t - t_* - v \cdot x(t)$. In the comparison region combine $|x| \leq R + |t|$ with $|t| \leq |t_*| + |v||x|$. Once compactness is proved, Lecture 8's signed-FTC argument applies without a compact graph deformation.

Answer. The relevant support satisfies $|x| \leq (R + |t_*|)/(1 - |v|)$. The graph pairing equals the flat one, and for the two specified solutions it is $\int f^2 \mathrm{d}^d x > 0$.

Exercise 8.2 (Common boundary conditions remove side flux). Let two real solutions of $(\partial_t^2 - \partial_x^2 + m^2)h = 0$ be smooth on a neighbourhood of $I \times [0, L]$, where I is an open interval, $L > 0$ and $m \geq 0$. Set $\Omega_{[0,L]}(h_1, h_2) = \int_0^L (h_1 h_{2,t} - h_2 h_{1,t})\mathrm{d}x$. At each endpoint, consider the following common conditions on both solutions: Dirichlet $h = 0$, Neumann $\partial_n h = 0$, or real Robin $\partial_n h + \kappa h = 0$, with the same prescribed real κ for both solutions at that endpoint. The two endpoints may have different such conditions.

(a) Prove that each choice makes the side symplectic flux zero and conserves $\Omega_{[0,L]}$. Specify the outward derivative at each endpoint.

(b) For $m = 0$, $L = \pi$, compare $h_D(t, x) = \sin x \cos t$ and $h_N(t, x) = 1$. Each obeys one of the listed conditions, but they obey different conditions. Compute their current, pairing and net outward side flux. Is individual satisfaction of some listed condition sufficient?

Hint. The outward side flux is the negative of $h_1\partial_n h_2 - h_2\partial_n h_1$. The cancellation requires the same endpoint relation for both arguments.

Answer. The common conditions give zero side flux and a constant pairing. For the mixed example, $\omega^0 = \sin x \sin t$, $\omega^1 = \cos x \cos t$, $\Omega = 2\sin t$, and the net outward flux is $-2\cos t$. Different conditions do not suffice.

Exercise 8.3 (A causal source bracket and its sign). In $3+1$ dimensions let G_{ret} be Lecture 6's massless kernel, let $G_{\mathrm{adv}}(t,x) = G_{\mathrm{ret}}(-t,x)$, and put $E = G_{\mathrm{adv}} - G_{\mathrm{ret}}$. For real $f, g \in C_c^\infty(\mathbb{R}^{1+3})$, define on the massless compact-data solution space

$$F_f(\phi) = \int_{\mathbb{R}^{1+3}} f\phi \, \mathrm{d}t\mathrm{d}^3x, \qquad Ef = (G_{\mathrm{adv}} - G_{\mathrm{ret}}) * f.$$

The identification of the resulting causal bracket with the linear *Peierls bracket* may be taken as given; see Khavkine, §§3.3.4–3.3.5, Theorem 3.2, and the scalar case §4.1, arXiv:1402.1282. The signs in this problem must be fixed by the following calculation, using the course convention $\iota_{X_F}\Omega = DF$.

(a) Show that Ef is a smooth homogeneous solution with compact smooth data on every constant slice. Apply the sourced current balance to $G_{\mathrm{ret}} * f$ and a homogeneous compact-data solution h to prove $\Omega(Ef, h) = \int fh$. Identify X_{F_f} and derive the bracket $\{F_f, F_g\} = \int f(Eg)$. Prove its antisymmetry and its vanishing when the two test supports are spacelike separated.

(b) As a separate finite-dimensional sign test, take one oscillator $\ddot{q} + \nu^2 q = 0$, $\nu \geq 0$, with $\{q(0), p(0)\} = 1$ and $p = \dot{q}$. Compute $\{q(t), q(s)\}$ and $\{q(t), p(s)\}$, using continuous limits at $\nu = 0$. Recover the equal-time canonical sign. Explain why this finite model does not make point evaluation of a field admissible in Lecture 7.

Hint. For $r_f = G_{\mathrm{ret}} * f$, the current with h has divergence $-hf$. Compare slices before and after the source; on the late slice $Ef = -r_f$. The advanced field vanishes there.

Answer. $X_{F_f} = Ef$ and $\{F_f, F_g\} = \int f(Eg)$. The oscillator brackets are $\sin(\nu(s-t))/\nu$ and $\cos(\nu(s-t))$, with limits $s - t$ and 1. In particular $\{q(t), p(t)\} = 1$.